BridgePulse-RL: Deep Q-Learning for Budget-Constrained Bridge Inspection and Repair

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Code, data, trained agents and results: https://github.com/b23bb1023/Bridge-RL-agent

I. Problem and environment

A road agency must keep ten aging bridges open for ten years (T=120 months) on half the money it needs, without seeing their true condition. Each month it issues one work order. The bridges are real Pennsylvania bridges from the National Bridge Inventory (NBI), with their deck areas, traffic (ADT), truck shares and detour lengths (3 arterial, 4 collector, 3 local). Funding of \$0.44M/yr, half the level at which the best tuned rule gets within 10% of its unlimited-money cost, accrues monthly into an account that carries savings forward.

Hidden dynamics, fitted to real data. Bridge i has condition $c_i \in \{0..4\}$ (Excellent to Critical, from its lowest NBI component rating). Without repair, it drops one state per month with probability $q_c \exp(\beta_a \frac{\text{age}_i - 50}{50} + \beta_w(w_i - \frac{1}{2}))$, where w_i is log traffic. The hazards q_c, the effects β_a=0.63 and β_w=−0.74, the routine rating noise (σ=0.35) and the hidden-state mix are fitted jointly by maximum likelihood. We use an HMM likelihood over 15,190 pairs of 2022/2024 inspections, so that rating noise is not mistaken for decay. Inventory bridges get upkeep we do not model (59–64 years per state), so the hazards are scaled by 3 (20–21 years). An open Critical bridge closes w.p. 0.25 per month until a major repair.

Observations and belief. An inspection reports $o \sim O(o|c) \propto e^{-(o-c)^2/2\sigma^2}$. Routine inspections are free and happen every 24 months with the fitted σ. Paid in-depth inspections (σ=0.25) can be bought any month. The environment keeps the exact posterior $b_i \in \Delta^4$ of every bridge. It is updated by Bayes' rule for inspections, by the repair and decay kernels, and by closures: a closure reveals c_i=4, and staying open is weak evidence against it. Initial conditions are drawn from the same posterior, so the filter is exact; unit tests check it by brute-force enumeration.

State $s_t \in [0,1]^{103}$: for each bridge, b_i, log closure and repair costs, age, months since inspection, and a closed flag; plus the account balance, month of year and t/T. **Actions** ($|\mathcal{A}|$=31): defer; inspect bridge i; minor repair of i (Good/Fair improve one state w.p. 0.9); major repair of i (to Excellent w.p. 0.95, else Good; reopens the bridge). A mask removes unaffordable work orders, and inspection or minor repair of a closed bridge. **Reward** (\$M): $r_t = -[C(a_t) + \sum_i D_i U_i + \sum_i (1-D_i)\rho(c_i)R_i + \mathbf{1}[t{=}T]\sum_i S_i(c_i)]$. C is the agency cost (16/160/1,600 \$/m^2 of deck for inspection, minor or major repair). U_i is the monthly road-user cost of a closure (D_i=1), ADT×detour×vehicle cost. ρ=[0, .05, .2, .6, 1] scales a structural-risk cost R_i. S_i is the cost to restore the bridge at the end. The negative return is the true 10-year cost, which we report. In training, the risk and restore terms are replaced by their belief expectation, which leaves expected returns unchanged and lowers the variance.

II. Agent and baselines

Decomposed masked DQN. A shared encoder embeds each bridge's features with the global ones. Mean/max pooling gives a network context. From its embedding and the context, bridge i outputs a baseline B_i, values Q_i^k for each work order k on itself, and a shadow price of money $\lambda_i \geq 0$. Then $Q(s,a) = \sum_i Q_i(s,a)$, where Q_i is B_i for defer, Q_i^k for work on i, and $B_i - \lambda_i \,\text{cost}_{kj}$ for work on another bridge j, which affects i only through the money it uses. Each Q_i is trained on bridge i's share of the reward, with targets at the joint greedy next action. The agent uses a masked $\arg\max$ when acting and in the double-DQN target, 3-step returns, γ=0.99, soft targets (τ=0.005), 8 environments and 3×10^5 steps. The checkpoint with the best cost on 40 validation episodes is kept.

Learning from a planner. A lookahead planner improves the best tuned rule by simulation. Each month it shortlists the rule's choice, defer, the four repairs saving the most risk per dollar, and the most informative inspection. It scores each candidate by 8 rollouts of that work order followed by the rule to the horizon. Rollouts start from conditions sampled from the belief (stratified per bridge) and share random numbers. The planner leaves the rule's choice only if the paired saving exceeds one standard error. We record 100 planner episodes and pre-train the DQN on them with TD plus a large-margin loss $\max_a[Q(s,a) + 0.5\,\mathbf{1}(a \neq a_E)] - Q(s,a_E)$. During Q-learning, a quarter of each batch comes from the demonstrations and the margin fades to zero.

Baselines and protocol. *Rules:* the four in Table I, with thresholds grid-searched on separate tuning episodes. *Maskable PPO* with 10^6 steps. *DQN ablations:* rule instead of planner demonstrations; none; an undecomposed shared network; a flat MLP; vanilla DQN (no masks, dueling, double estimator or demonstrations). *Information variants:* the agent sees the truth, only the last inspection, or cannot buy inspections, each warm-started by a planner with the same information. The main agent and PPO use 5 seeds, the ablations 2. The test uses 200 held-out episodes on which every policy faces the same random draws, so differences are paired, with 95% bootstrap CIs.

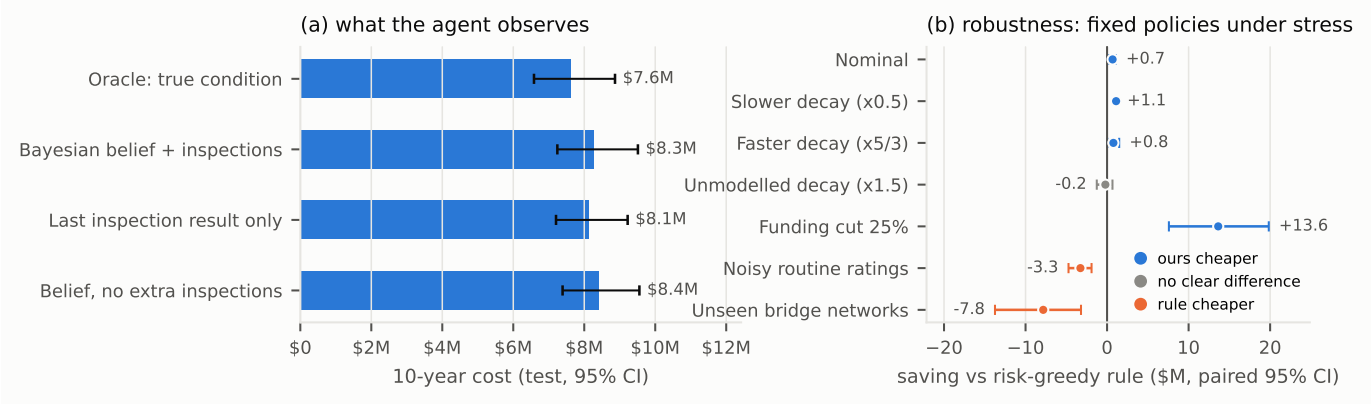


Fig. 1. (a) Test cost by what the agent observes (95% CI). (b) Paired saving of our agent over the best tuned rule when the world differs from training (no retraining; 95% CI).

TABLE I
TEN-YEAR COST ON 200 TEST EPISODES ($M, 95% CI). Δ: PAIRED EXTRA COST OVER OUR AGENT. CLOSED: BRIDGE-MONTHS. INSP.: PAID INSPECTIONS. †SEES THE TRUE CONDITION.

Policy	Cost $M [95% CI]	Δ vs ours	Closed	Insp.
DQN (ours)	8.28 [7.24, 9.51]	–	6	1.7
DQN (ours), 5-seed ensemble	8.06 [7.09, 9.16]	−0.22	5	0.7
Lookahead planner (teacher)	8.61 [7.61, 9.82]	+0.33	6	4.2
DQN, offline pre-training only	8.26 [7.28, 9.35]	−0.02	6	1.0
DQN, rule demonstrations	8.96 [8.11, 9.99]	+0.68	5	0.0
DQN, no demonstrations	13.47 [12.32, 14.77]	+5.19	3	44.6
DQN, shared (undecomposed) net	8.76 [7.47, 10.26]	+0.48	6	0.6
DQN, flat MLP	10.84 [9.06, 12.85]	+2.56	10	2.0
Maskable PPO	14.93 [13.11, 16.93]	+6.64	12	78.1
Vanilla DQN (no mask)	159 [137, 182]	+151	88	71.3
DQN, last inspection only	8.14 [7.20, 9.22]	−0.14	5	0.5
DQN, no extra inspections	8.41 [7.39, 9.55]	+0.13	6	0.0
DQN, true condition†	7.61 [6.58, 8.87]	−0.67	5	0.1
Rule: condition-based	9.37 [8.38, 10.56]	+1.08	6	0.0
Rule: entropy-triggered	12.20 [10.60, 14.04]	+3.92	13	1.1
Rule: risk-greedy	8.96 [8.11, 9.96]	+0.68	5	0.0
Rule: oracle threshold†	8.69 [7.78, 9.76]	+0.41	6	0.0
DQN, untrained	301 [253, 351]	+292	106	40.0
Do nothing	301 [253, 351]	+292	106	0.0

III. RESULTS

Main result. Over five seeds, the DQN's 10-year cost is $8.28M [7.24, 9.51]. That is 8% below the best tuned rule (risk-greedy, $8.96M), a paired saving of $0.68M [0.21, 1.08] (Table I). Averaging the five networks' Q-values gives $8.06M, $0.22M [0.07, 0.42] better again. It matches its teacher: it is $0.33M [−0.21, 1.02] cheaper than the planner. It needs 0.62 ms per decision instead of 749 ms. With about as many repairs as the rule, it spends $2.87M instead of $2.37M. It cuts structural risk from $4.44M to $3.33M and the final restore cost from $0.68M to $0.12M, at slightly more closure (5.7 vs 4.8 bridge-months).

What the gain comes from. The gain is learned offline. Q-learning with the margin on the planner's episodes alone gives $8.26M, within noise of the full agent. Online Q-learning holds that level while the margin lasts, and every seed degrades once it reaches zero (Fig. 2); the best validation checkpoint is kept. The teacher matters: with rule demonstrations the agent only matches the rule ($8.96M), and from scratch it costs

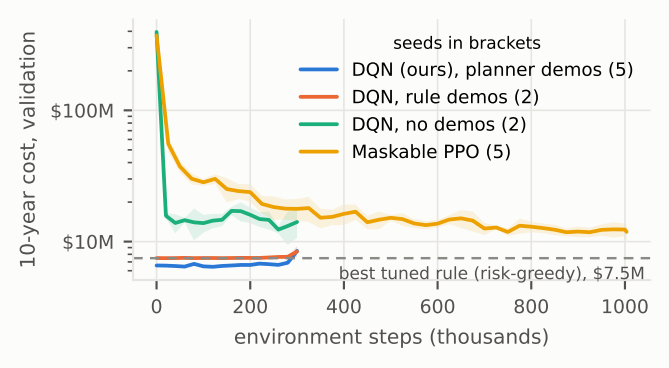


Fig. 2. Validation cost during training (mean, 95% CI over seeds). Step 0 is after offline pre-training on the demonstrations; the jump at the end is where the margin reaches zero.

$13.47M. The decomposition matters too: the undecomposed shared network costs $8.76M (+0.48 [0.14, 0.88]) and the flat MLP $10.84M. Maskable PPO ($14.93M) and vanilla DQN ($159M) do not reach the rules.

Value of information (Fig. 1a). Seeing the true condition is worth $0.67M [0.31, 1.02]. Neither the belief nor paid inspections recover much of it: agents that see only the last inspection ($8.14M) or cannot buy inspections ($8.41M) are within noise of ours. Routine ratings are accurate here (σ=0.35), so the last rating and its age carry most of what the posterior knows.

Robustness (Fig. 1b). Without retraining, the agent stays cheaper than the best rule under slower decay (+1.11) and faster decay (+0.77), and by far under a 25% funding cut ($15.05M vs $28.7M). It is no better when decay is faster than its model expects (−0.21 [−1.26, 0.66]). It is worse with noisier routine ratings ($15.70M vs $12.43M) and on twenty unseen networks of real bridges ($21.0M vs $13.13M). A policy trained on one network does not transfer, while the rules do by construction.

Limitations. The agent is trained for one network (training across sampled networks is the next step); unit costs, the risk rate and the unmaintained factor are assumptions.